

SWARAJ KUMAR SAHOO

Jajpur Road, Jajpur, Odisha

📞 7205762913 ✉️ sswarajkumar848@gmail.com 🌐 swarajsahoo 🎮 Jaraws12

Education

National Institute of Technology, Rourkela

INTEGRATED MSc MATHEMATICS (CGPA: 7.89)

November 2022 – Present

Rourkela, Odisha

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Deep Learning
- Database Management
- Machine Learning
- Probability & Statistics

Experience

Eunora Technology

May 2026 – July 2026

Software Engineer Intern

Remote

- Developed a cross-platform React Native (Expo) application for an agricultural services platform with secure OTP-based user authentication, persistent login using AsyncStorage, and protected navigation using Expo Router.
- Integrated a Firebase Firestore backend to manage real-time service bookings, enabling users to create, retrieve, and track booking history through user-specific database queries and filtering.
- Built responsive mobile interfaces with reusable components, multilingual support, and booking management workflows, connected to backend REST APIs to support the full booking flow.

Projects

Accident-GIS | *Machine Learning, APIs, LangChain, React*

May 2026

GitHub

- Engineered a FastAPI backend exposing 3+ REST endpoints for environmental data ingestion, risk scoring, and real-time prediction serving.
- Implemented a full-stack ML inference pipeline, training and deploying 1 machine learning model for real-time frontend-backend communication.
- Designed an interactive GIS dashboard with Leaflet featuring real-time location tracking, heatmap visualization, AI risk indicators, and weather analytics for accident-prone areas.
- Automated data collection by integrating 3 external services (Geolocation API, OpenWeather API, and OpenStreetMap) to generate safety recommendations from real-time environmental conditions.

Knowledge Assistant | *FastAPI, React.js, LangChain, FAISS*

June 2026

GitHub

- Created a full-stack Knowledge Assistant using React, FastAPI, FAISS, SQLite, and Gemini 2.5 Flash, enabling semantic search and conversational querying over 100+ page PDF documents with real-time streaming responses.
- Constructed a hybrid Retrieval-Augmented Generation (RAG) pipeline combining FAISS vector search, BM25 lexical retrieval, and cross-encoder reranking across 3 retrieval stages to improve answer relevance.
- Streamlined a document ingestion pipeline with semantic chunking, metadata indexing, multi-document filtering, and source attribution, supporting retrieval across thousands of indexed chunks.
- Architected a backend with 20+ REST API endpoints for document management, chat persistence, AI-powered conversations, and streaming responses.

Achievements & Certifications

- **April Cohort Hackathon (April 2025): Winner**
- **SBI Ideathon (February 2025): Top 1000 out of 50000 teams**
- **Hack4Bengal 4.0 (June 2025): Top 100 out of 1500 teams**

Technical Skills

- **Languages:** C++, Python, JavaScript, SQL, HTML, CSS
- **Developer Tools:** Git, GitHub, Postman, VS Code, Google Colab
- **Frameworks/Libraries:** React.js, Node.js, Express.js, FastAPI, Tailwind CSS, Bootstrap, Redux, Mongoose, Material UI, LangChain, FAISS
- **AI/ML:** Gemini API, FAISS, BM25, Hugging Face Transformers, Scikit-learn, TensorFlow, OpenCV

Extracurricular Activities

AXIOM, Math Club – NIT Rourkela

May 2024 – May 2025

Secretary

Algorithmic & Programming Society – NIT Rourkela

August 2024 – May 2025

Event Management Lead